

An element and the cells around it: OEIS A277768

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7 September 2026

Abstract

OEIS A277768 carries an empirical recurrence of order 18 contributed by R. H. Hardin. It is true, and it is decidable rather than empirical. The entry's condition is decided by a bounded amount of state carried down the array; the admissible windows are the vertices of a finite digraph and the arrays counted are exactly the walks in it. Hence $a(n)$ is a walk count on $S = 23$ vertices and is C -finite, and whether the conjectured recurrence annihilates it is settled by a finite exact computation, with the Cayley–Hamilton theorem bounding the work.

2020 Mathematics Subject Classification. 05A15, 05B45, 15B36.

1 The conjecture

OEIS A277768 is “Number of $3 \times n$ 0..2 arrays with every element equal to some element at offset $(-1,-1)$ $(-1,0)$ $(-1,1)$ $(0,-1)$ $(0,1)$ or $(1,0)$ both plus 1 mod 3 and minus 1 mod 3, with new values introduced in order 0..2.”. It has offset 1 and begins

$$0, 2, 17, 204, 1989, 21141, 220549, 2292380, \dots$$

The entry states, as a conjecture contributed by its author and never marked settled:

$$\begin{aligned} \text{Empirical: } a(n) = & 11*a(n-1) - a(n-2) + 10*a(n-3) - 648*a(n-4) - 323*a(n-5) + 2443*a(n-6) \\ & + 9234*a(n-7) - 2121*a(n-8) + 1974*a(n-9) - 6288*a(n-10) + 2781*a(n-11) - 4959*a(n-12) \\ & + 757*a(n-13) - 898*a(n-14) + 2180*a(n-15) + 116*a(n-16) + 44*a(n-17) - 288*a(n-18) \end{aligned}$$

As of the “Last modified” line on the live entry (Oct 29 2016 07:31:30, revision 4) this is still recorded as empirical, and nothing on the entry records it as proved.

2 The count is a walk count

The entry counts arrays with 3 rows over the alphabet $\{0, 1, 2\}$, growing one column at a time, and asks that every element x must have, among the cells at the offsets $(-1, -1)$, $(-1, 0)$, $(-1, 1)$, $(0, -1)$, $(0, 1)$, $(1, 0)$, one equal to $x + 1$ and one equal to $x - 1$, both modulo 3.

The furthest that condition reaches is 1 column, so a window of 3 consecutive columns decides it for the column in the middle. Carry 2 columns as the state; a column is judged when the column 1 beyond it arrives, and the last 1 column — the ones with nothing far enough beyond them — are judged by the end vector. A neighbour off the edge of the array is not a neighbour, which is what the entry means and what keeps the first few terms small.

The entry also asks that new values be introduced in row major order. That is not a condition on any window — whether a value may appear here depends on the whole prefix — but it enters through a single integer, how many values have been introduced so far, and the effect is that

each array is counted once for the whole class of arrays differing from it only by renaming the values.

Merging vertices with identical futures leaves $S = 23$. An admissible array is exactly a walk, so with M the adjacency matrix and ι, τ the vectors recording which states may begin and end an array,

$$a(n) = \iota^\top M^{n-1} \tau,$$

the exponent fixed by the entry's own shape and checked against its published terms. In particular a is C -finite.

3 The proof

Lemma 1. *Let $q(t) = t^{18} - \sum_i c_i t^{18-i}$ be the characteristic polynomial of the conjectured recurrence. The residual $a(n) - \sum_i c_i a(n-i)$ equals $\iota^\top M^j q(M) \tau$ for the corresponding walk index j , so the recurrence holds from some point on if and only if $\iota^\top M^j q(M) \tau = 0$ for all large j , and it is enough to examine $j < S$.*

Proof. Factor M^j out of $M^{j+18} - \sum_i c_i M^{j+18-i}$. For the bound, M^S is an integer combination of $I, M, \dots, M^{S-1}$ by Cayley–Hamilton, so the numbers $u_j = \iota^\top M^j q(M) \tau$ obey the monic recurrence given by the characteristic polynomial of M ; S consecutive zeros force every later one, and the last nonzero u_j pins the threshold exactly. A constant factor in front of a divides out of the residual and changes nothing here. $\square$

Theorem 2.

$$a(n) = 11a(n-1) - a(n-2) + 10a(n-3) - 648a(n-4) - 323a(n-5) + 2443a(n-6) + 9234a(n-7) - 2121a(n-8) + 1$$

for every $n > 18$.

Proof. The vector $w = q(M)\tau$ was formed by 18 matrix–vector products in exact integer arithmetic and $\iota^\top M^j w$ evaluated for $j = 0, 1, \dots$, again exactly; those integers vanish from the index corresponding to $n = 18 + 1$ onwards and the run continued until the working vector was identically zero, which settles every larger j at once. $\square$

4 Verification

Four checks, each independent of the algebra above.

First, the model reproduces all 19 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry: the digraph is built by reading the entry's English, and a misreading gives different counts.

Second, the reading was pinned before the digraph was written, by a program that enumerates every array of the given shape one by one and tests the entry's condition at each cell directly. That program shares no code with the transfer matrix, so an error in one does not hide an error in the other.

Third, the conjectured recurrence was evaluated directly on the published terms, with no matrices involved, and holds wherever the proved range and the data overlap.

Fourth, the annihilation test of Lemma 1 was carried out in exact integer arithmetic on the whole vector rather than on a sampled prefix.

References

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